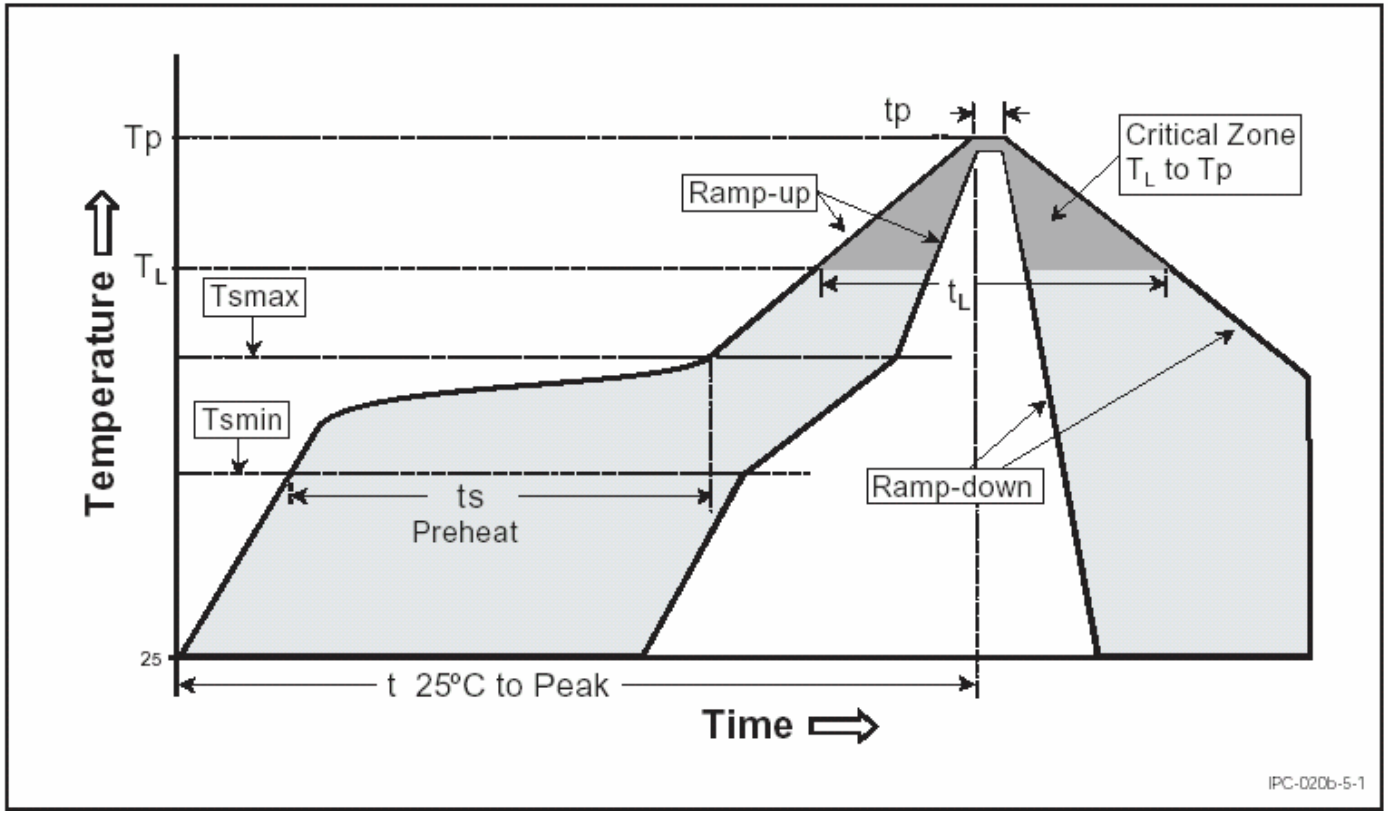
General Reflow Profiles

Recommend Reflow Profile

A PCBA reflow profile depends on the thermal mass of the entire populated board. The actual temperature used in the reflow oven is a function of:

- Solder paste types
- Board density
- Component location
- Component mass
- Board finish



General Reflow Profiles

Profile Feature	Sn-Pb Eutectic Assembly		Pb-Free Assembly	
	Large Body	Small Body	Large Body	Small Body
Average ramp-up rate (T_{smax} to T_p)	2℃/second max		2℃/second max	
Preheat -Temperature Min (Ts_{min}) -Temperature Max (Ts_{max}) -Time (min to max) (ts)	100℃ 150℃ 60-120 seconds		150℃ 200℃ 60-180 seconds	
Time maintained above: -Temperature (T_L) -Time (t_L)	183℃ 60-150 seconds		217℃ 60-150 seconds	
Peak Temperature (Tp)	225+0/-5℃		245+5/-5℃	
Time within 5℃ of actual Peak Temperature (tp)	20 seconds max		30 seconds max	
Ramp-down Rate	3℃/second max.		3℃/second max.	
Time 25℃ to Peak Temperature	6 minutes max.		8 minutes max.	

Note:

1.All temperatures refer to topside of the package, measured on the package body surface.

2.Actual board assembly depends on other parts on board density and follower solder paste manufacturers's guideline.